

MODERN DATA ANALYTICS

Case Study: Exploring Unusual Patterns in Cycling Traffic in Flanders

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Group Number: 33

Anna Solodka (r0926269)

Nerissa Truong (r0823591)

Yuxi Liu (r1070939)

Maria Amelie Gonda (r0910823)

1 Introduction

In countries where cycling is one of the main means of transportation, cycling traffic follows regular patterns across time and locations. A large share of trips is undertaken for routine purposes and takes place at fixed times of the day over many months and years. For a given time and location, it is therefore possible to define an expected level of cycling activity. Observed counts, however, do not always coincide with this expectation. Such differences are not necessarily isolated and may themselves exhibit a degree of regularity. It is that regularity that the present study is intended to explore.

In Flanders, such an exploration can be carried out using automatic bicycle count data collected by the Agentschap Wegen en Verkeer¹. These data consist of cyclist counts recorded at fixed monitoring sites, where each observation is defined by a specific time interval, location, and direction of movement. Since the observations are repeated over time, they provide a basis for describing what is normally recorded under comparable conditions. The analysis therefore begins with the estimation of this normal level of traffic, against which observed counts are then compared. From this comparison, deviations are defined. A deviation is not defined by any difference between the observed and expected count. In count data, some difference is expected. The term deviation is thus reserved for differences large enough to exceed ordinary short-term variation in cycling traffic counts.

The occurrence of these deviations is then assessed in relation to external factors that may disturb regular traffic patterns. Weather, major cultural events, mass cycling activities and transport-related disruptions are considered for this purpose. The analysis concerns both the association between these factors and deviations, and the extent to which this association differs between sites. Some sites may be prone to deviations as such, whereas others may be prone to deviations mainly when one or more external factors are present. The ability to make that distinction is important for infrastructure monitoring and targeted intervention planning. It determines whether intervention should address the site’s general capacity to accommodate irregular traffic, or its capacity to do so in the presence of the factors with which most deviations are associated.

2 Data Acquisition and Preprocessing

The analysis uses AWW traffic count data for the period from May 2022 to April 2026. The raw count data are provided as separate monthly CSV files, available from the AWW open data repository. Each observation reports the number of road users passing one of the automated counting sites during a 15-minute interval and specifies a direction of movement. The corresponding site metadata, including the geographical locations of the counting points, were obtained from the same source.

Several external datasets were acquired to provide context for the traffic observations. Public holiday dates for Belgium were obtained from the Nager.Date public holidays API. School holiday periods were obtained from the OpenHolidays API and restricted to the Flemish school system. These variables account for regular changes in cycling activity associated with the calendar and are therefore used to define the expected normal level of traffic. For the same expected-count modelling framework, fuel price data were obtained from the European Commission oil bulletin.

A dataset containing information on weather, as one of the factors which may account for deviations in cycling counts, was obtained from the Open-Meteo API. Hourly weather data included temperature, precipitation, rain, and snowfall. Information on major cultural and

¹Agentschap Wegen en Verkeer is the Flemish government agency responsible for managing, maintaining, and developing the regional road network and related transport infrastructure in Flanders.

sport events potentially capable of causing deviations in cycling traffic, as well as transport-related strikes, was collected manually from several event calendars and news sources due to the lack of complete datasets for these factors.² For this analysis to be replicable, the file with the collected events was deposited in the GitHub repository.

Before the analysis could be conducted, the raw count files had to be brought into a form suitable for comparison across months, sites, and directions. This was executed in Python using the pandas package. The monthly count files were first read using a common set of column names. The site and direction metadata files were read separately. Date variables were converted to datetime format, and the count variable was converted to numeric form.

The count data were then restricted to records for which the traffic type was FIETSERS, that is, cyclists. The duration of each recorded interval was subsequently examined. For almost all observations, the recorded intervals were equal to the expected 15-minute period. A small number of observations, however, had non-standard interval lengths due to the spring daylight-saving time change. The observations associated with those non-standard intervals were removed from the analysis. Their removal is unlikely to bias the results, given that the share of those observations was only approximately 0.0007%.

Since the 15-minute intervals are highly susceptible to short-term fluctuations in the recorded counts, they were not used as the main unit of analysis. The 15-minute counts were instead aggregated to 2-hour intervals. These intervals are short enough to preserve variation within the day, but long enough for short-term changes in counts to average out. The aggregation was defined by site, direction, date, month, weekday, and 2-hour time bin. For each aggregated row, the total cyclist count was computed, together with the number of available 15-minute count observations and the number of missing count observations. Intervals with at least 5 out of the expected 8 observations were retained for modelling. For these mostly complete intervals, an adjusted count was computed by scaling the observed sum to an 8-interval equivalent. Intervals with fewer than 5 observed observations were excluded from the modelling dataset. This approach to dealing with missing values resulted in the removal of 2.5% of the aggregated rows.

After aggregation, site metadata were added to each observation. This included the geographical coordinates, site name, municipality, district, and installation date of the sensor. Direction metadata were also added, so that each observation retained both the coded direction and the textual description of that direction. School holiday and public holiday data, with the name and binary indicator, were then matched to the data by date. Fuel prices were available at weekly frequency. For this reason, the fuel-price series was first converted to a daily series. Each calendar day was assigned the most recent available petrol price. The daily fuel-price variable was then merged with the aggregated count data by date.

From the combined dataset, the period from 1 May 2025 to 30 April 2026 was then separated. This period was used as the prediction period, for which expected counts were estimated and deviations were later defined. Sites for which data were present only in this prediction period were removed from this dataset to avoid producing expected counts for locations not observed during the model-development period. It is to this dataset that the data on external factors were added.

Weather data were first aggregated from hourly records to 2-hour intervals. For each 2-hour interval, mean temperature was computed, and values for rain and snow precipitation were summed. The precipitation variable was later categorised, with five categories defined: dry, light precipitation, moderate precipitation, heavy precipitation, and snow. Created variables

²The sources included Visit Flanders and city event calendars for cultural events, The Brussels Times, VRT NWS, and Strike Tracker Belgium for transport strikes, and cycling or sport calendars such as Cycling in Flanders, UCI, Ronde van Vlaanderen, and We Ride Flanders for sport events.

were then matched to the prediction-period observations by site, date, and 2-hour time bin. The event data were also added. Transport strikes were matched by date and by the geographical scope of the strike. Outdoor and indoor music events were matched by date, location, and event time. Sport events were matched by date and location.

The preprocessing therefore produced two main analysis datasets. The first dataset contains the cleaned and aggregated count data with site, calendar, and fuel price variables for the time period from May 2022 to April 2026, later used for model development. The second contains only prediction-period observations with the additional weather and event variables, and is used for defining and explaining deviations from expected counts.

3 Exploratory Data Analysis

3.1 Full Modelling Dataset

The exploratory analysis was first conducted for the full modelling dataset. This dataset contains 4,587,736 observations and has no missing values in the variables used for normal-count modelling. The observations cover two recorded directions, IN and OUT, 12 two-hour time bins, 7 weekdays, and 12 months.

The observations are almost evenly distributed across the variables that define repeated time structure. Each 2-hour bin accounts for about 8.3% of the data, and each weekday for about 14.3%. Month shares are not exactly equal, which is expected because months differ in length, but they remain close enough for all months to be represented in the data. The direction variable is also almost evenly split between IN and OUT.

Other variables are less balanced by construction. Public holidays account for 3.8% of the observations, while school holidays account for 28.5%. This difference does not indicate a problem with the data, but follows from the calendar itself. Fuel prices are complete after matching to the count data, with most values concentrated between 1.550 and 1.72 euro per litre.

The response variable used for normal-count modelling is the rescaled cyclist count. Its distribution is strongly right-skewed (see Figure 1). The mean count is 21.36, while the median is 4. The 75th percentile is 21, the 95th percentile is 96, and the 99th percentile is 235. The maximum value is 8,682. Thus, most observations correspond to low or moderate traffic, while a small part of the data contains very large counts.

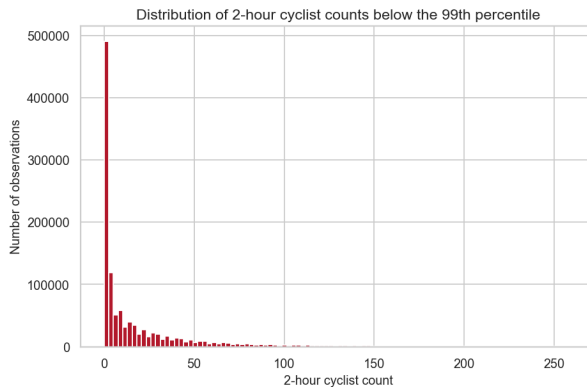


Figure 1: Distribution of 2-hour cyclist counts below the 99th percentile

Zero counts are also a substantial part of the response distribution. In total, 1,254,674 observations are equal to zero, corresponding to 27.35% of the sample. The response variable therefore

has two features that are important for the modelling step. First, the long upper tail suggests that the variance may be much larger than the mean, so overdispersion has to be checked if a Poisson model is used. Second, the large share of zero counts means that the selected model should not only predict average traffic levels well, but should also represent low-traffic and zero-traffic periods with reasonable accuracy.

3.2 Prediction-Period Dataset with External Factors

As mentioned in the data preprocessing section, the expanded dataset was built upon the original one with a shorter date range (May 2025–April 2026) to explore the deviations from the regular, everyday rhythm of bicycle traffic in Flanders. After preprocessing, the dataset contains roughly 1.17 million observations and introduces some external factors, specifically weather metrics and scheduled events, which are designed to help us capture unsystematic shocks to the cycling network.

The distribution of the rescaled cyclist count in this period is similar to that observed in the full modelling dataset. The median count is 5, compared with 4 in the full dataset, and 25.6% of observations are equal to zero, compared with 27.35% in the full dataset. Thus, the prediction period does not appear to differ significantly from the earlier data in its basic count distribution.

The prediction-period data are also similarly distributed across the repeated time variables. The hour bins, weekdays, months, and directions are all well represented. Public holidays and school holidays occur in the proportions implied by the calendar, while site coverage remains uneven. This unevenness is expected, since not all counting sites are active or complete for the same length of time.

Weather variables were examined at the same 2-hour level as the count data. Temperature is mostly concentrated between 5°C and 20°C, with many observations between 10°C and 15°C (see Figure 2a). Rain and snow variables are concentrated at zero. While this might seem unusual for Belgium at first glance, it perfectly reflects the nature of 2-hour interval data: most specific 2-hour windows in a year are dry. When looking at the categorical breakdown, roughly 1 out of every 5 intervals (about 20.3%) actually experienced some form of precipitation, either light, moderate, heavy, or snow (see Figure 2b).

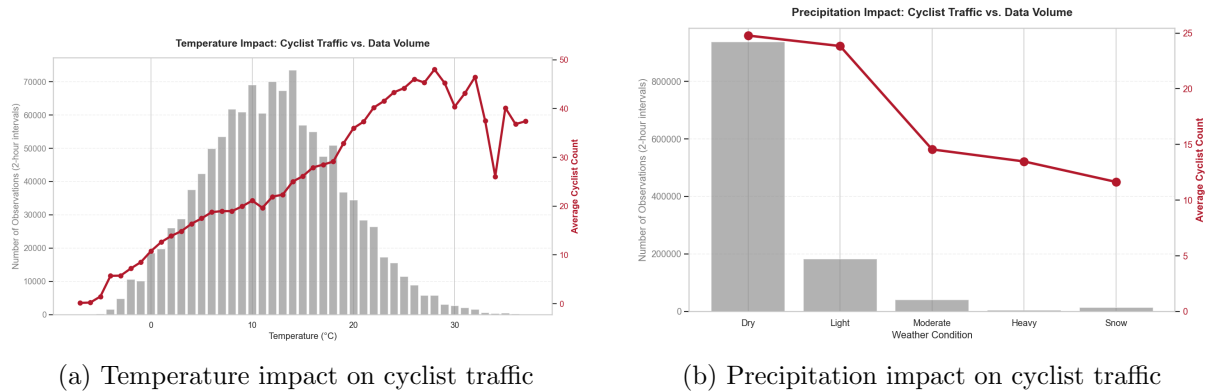


Figure 2: Impact of weather on cyclist traffic

Taking a closer look to compare the raw averages alongside data volume, two distinct patterns emerge. A clear upward trend appears in the raw data, suggesting a positive relationship between average cyclist counts and hot weather. A predictable downward trend also appears as precipitation conditions worsen, with average counts dropping sequentially from dry, to light precipitation, moderate precipitation, heavy precipitation, and finally snow. These comparisons

are, however, based on absolute cyclist counts rather than deviations from expected counts. They therefore do not yet show whether weather is associated with deviations. They only describe how observed counts differ across weather conditions before the expected-count baseline is taken into account.

Unlike weather, which is observed for all intervals, scheduled events are rare and episodic. The exploratory analysis shows that variables such as strikes, music events, and sporting events are highly imbalanced. This is expected, since such events are restricted to particular dates, locations, and, in some cases, times of day. Strikes are the most frequent of these variables, occurring in about 4.1% of the observed intervals. Outdoor music events, indoor concerts, and sporting events, such as cyclocross or road cycling races, are much more localised, appearing in less than 0.2% of the dataset. Because these events are rare, modelling their association with deviations will rely heavily on the geographic and temporal matching established during the data merge.

For the moment, a simple relative difference in cyclist counts between event and non-event intervals was calculated to describe their preliminary pattern. The raw data show higher traffic during indoor music events, with an increase of more than 120% compared to non-event intervals, followed by outdoor music events, with an increase of nearly 50%, and sport events, with an increase of about 20% (see Figure 3). The difference on strike dates is smaller, around 6%, but is based on a larger share of observations. These are still raw comparisons of observed counts, not comparisons of deviations from expected counts.

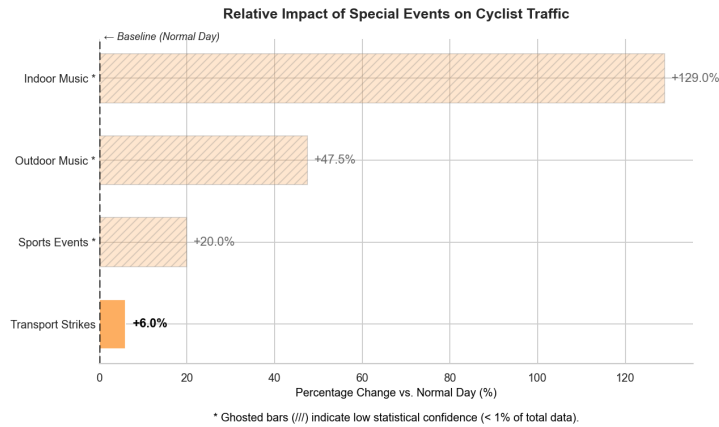


Figure 3: Relative impact of special events on cyclist traffic

Overall, the prediction-period dataset retains the main count structure observed in the full modelling dataset. The weather variables describe ordinary variation in external conditions, while the event indicators identify selected rare and localised disruptions. The raw differences described above are therefore treated only as preliminary observations, to be examined later after deviations from expected counts are defined.

4 Modelling Normal Cycling Patterns

The later definition of a deviation is based on the difference between the observed number of cyclists and the expected normal number of cyclists. Thus, for deviations to be defined, the expected number of cyclists must first be estimated.

The modelling of expected cycling counts was treated as a supervised regression problem organised around the regular structure of the observed cyclist counts per 2-hour interval. This

structure is the part of the observed variation that can be explained by temporal, spatial, directional, calendar-related, and broader background variation.

$$\text{Observed cyclist count} = \text{Regular structure} + \text{Irregular deviation} + \text{Noise} \quad (1)$$

Fuel price was included as one such background covariate. Its function in the model was to account for gradual changes in the level of cycling traffic that may be related to the cost of motorised transport. This differs from external factors such as events, strikes, or infrastructure works, which are localised in time or space and are considered after deviations have been defined.

Within this supervised framework, both likelihood-based count models and a machine-learning regression model suitable for estimating non-negative discrete counts were fitted. The estimation period covers data recorded between May 2022 and April 2025. This period was split into two parts. The first part, from May 2022 to April 2024, was used for the training purposes and earlier model diagnostics. The second part, from May 2024 to April 2025, was used as a validation sample to evaluate how well the models perform on data that was not used during estimation.

4.1 Poisson Regression Model

The first model considered was a Poisson regression model, fitted with the standard log link, using the statsmodels package. The model was specified as follows:

$$\begin{aligned} \log(\text{Expected Count}) = & \beta_0 + \beta_1 \text{Hour} + \beta_2 \text{Weekday} + \beta_3 \text{Month} \\ & + \beta_4 \text{PublicHoliday} + \beta_5 \text{SchoolHoliday} + \beta_6 \text{Site} \\ & + \beta_7 \text{Direction} + \beta_8 \text{FuelPrice} + \log(\text{Number of Periods}) \end{aligned} \quad (2)$$

There were a number of software constraints that determined this specification. Since the data used have a longitudinal structure, with observations for the same counting sites recorded repeatedly over time, part of the variation in the counts is site-bound. In longitudinal data analysis, it is often preferable to use a mixed-effects model with random effects for the repeatedly observed units, in order to account for unit-level variation. However, the Python packages used for frequentist generalised linear modelling do not provide a straightforward mixed-effects extension for the type of count model used here. Since not accounting for site-level variation would likely invalidate expected normal count estimations, the site variable was included as a categorical predictor.

The inclusion of site as a categorical predictor meant that the model had to estimate more than one hundred site coefficients. With this number of coefficients and more than two million row-level observations, direct estimation was computationally difficult. To reduce the size of the estimation problem, the training data were aggregated over the combinations of predictors used in the model. For this aggregation, fuel prices were rounded to one decimal place. For each aggregated row, the total cyclist count and the number of original 2-hour periods were recorded.

The aggregation does not change the count-model interpretation. If one group contains n_g periods, and the expected count for one period is μ_g , then the expected total count for that group is $n_g \times \mu_g$. On the log scale, this becomes:

$$\log(\text{Expected Total Count}) = \log(n_g) + \log(\mu_g) \quad (3)$$

The term $\log(n_g)$ was therefore included as an offset term to adjust the total count for the number of periods included in the aggregated row. The model is thus fitted to total counts, but the estimated mean still has the interpretation of an expected count per 2-hour period.

In the fitted Poisson model, all estimated coefficients were statistically significant. Yet, it needs to be taken into account that, with this sample size, even small differences from the reference categories can produce very small p-values. What is more important for the final model choice is whether the assumptions associated with the chosen model hold. For the Poisson model to be acceptable, the mean-variance assumption should be satisfied. This assumption states that the conditional variance of the count is equal to its conditional mean.

The diagnostics showed that this assumption is strongly violated. The Pearson dispersion was 52.60, and the deviance dispersion was 40.85. When the assumption is satisfied, these values should be close to 1. Another important aspect of the model diagnostics is the ability of the model to represent zero counts, since in the observed count data the share of zero counts is approximately 28.26%. The Poisson model strongly underpredicts zero counts, with the predicted share being only around 8.17%.

4.2 Negative Binomial Model

To account for overdispersion, a negative binomial model was fitted. It keeps the same log-linear mean structure, but includes a dispersion parameter, so that the variance is no longer forced to be equal to the mean.

The negative binomial model was also fitted on the aggregated data for the computations to be feasible. Here, however, one important limitation needs to be acknowledged. For the Poisson model, the sum of independent Poisson counts with the same covariate pattern is again Poisson, with mean equal to the sum of the individual means. For the negative binomial model, this property does not generally hold. The offset preserves the expected total count, but the distribution of the summed count is not necessarily the same as for the unaggregated counts. This means that the estimates obtained from the aggregated data are approximations to those that would be obtained from the original data.

For the negative binomial model, the same covariate structure as for the Poisson model was first used. The model was then extended to include two interactions for which there was a clear rationale. The hour-by-weekday interaction was included because the daily pattern of cycling is not expected to be the same across working days and weekends. The hour-by-direction interaction was included because directional flows may also change over the day, for example between morning and evening traffic.

$$\begin{aligned} \log(\text{Expected Total Count}) = & \beta_0 + \beta_1 \text{Hour} + \beta_2 \text{Weekday} + \beta_3 \text{Month} \\ & + \beta_4 \text{PublicHoliday} + \beta_5 \text{SchoolHoliday} + \beta_6 \text{Site} \\ & + \beta_7 \text{Direction} + \beta_8 \text{FuelPrice} \\ & + \beta_9 (\text{Hour} \times \text{Weekday}) + \beta_{10} (\text{Hour} \times \text{Direction}) \\ & + \log(\text{Number of Periods}) \end{aligned} \tag{4}$$

Indeed, the model with both interaction terms performed better than the baseline negative binomial model. The likelihood-ratio statistic for this comparison was 60,892.73 with 77 degrees of freedom and a p-value less than 0.001. The decrease in AIC was noticeable, from 6,792,270 to 6,731,531. Since both criteria changed in the same direction, the model with both interactions was used for further diagnostics and comparison.

The diagnostic assessment showed that, for the data modelled, the negative binomial model is more appropriate than the Poisson model. The estimated dispersion parameter was 1.54, which indicates that the negative binomial model did not reduce to the Poisson model. The implied zero-count share was 25.02%, which is much closer to the observed zero-count share of 28.26%.

4.3 Comparing Negative Binomial Model and Histogram Gradient Boosting Model

To assess whether a more flexible method would provide better predictive performance, the histogram gradient boosting method was used. It was implemented as a pipeline using the scikit-learn package. In the preprocessing part of the pipeline, categorical variables — site, direction, month, weekday, time interval, and holiday-related variables — were transformed into indicator variables using OneHotEncoder to prevent them from being treated as ordinal. The numeric fuel-price variable was kept unchanged.

The model was trained using Poisson deviance as the error criterion, which is suitable for non-negative count outcomes. Given the computational constraints, parameter tuning was done manually and only for a limited number of model specifications. The tested specifications differed in learning rate, maximum number of leaf nodes, and minimum number of observations per leaf. The best-performing specification used a learning rate of 0.05, a maximum of 31 leaf nodes, and a minimum of 50 observations per leaf.

The predictive performance of the negative binomial model and the histogram gradient boosting model was compared on the validation dataset. The histogram gradient boosting model had better validation performance in terms of mean absolute error (MAE), root mean squared error (RMSE), and mean Poisson deviance (see Table 1).

Table 1: Validation performance of the negative binomial and histogram gradient boosting models

Model	MAE	RMSE	Poisson deviance	Observed mean count	Predicted mean count
Negative binomial model	16.5161	39.8127	21.8443	22.5366	22.2970
Histogram gradient boosting model	12.3838	33.6298	14.9244	22.5366	18.8975

However, it systematically underpredicted average counts, both for the overall average and for the averages by all weekdays and most daytime hours. It also strongly underpredicted the implied share of zero counts in the validation dataset. The observed share was 26.83%, while the Poisson-implied zero-count share from the HGB predicted means was 6.13%.

The negative binomial model had larger pointwise errors, but its grouped mean predictions were generally closer to the observed validation means. Since the expected-count model was used as the baseline for later deviation detection, preserving the mean level and distributional structure of the counts was more important than minimising pointwise errors alone. For this reason, the negative binomial model was retained as the final expected-count model.

The model was refitted on the aggregated data for the full model-development period, from May 2022 to April 2025. This was done because the training-validation split was used for model choice. Once the model choice had been made, both periods could be used to estimate the final expected-count model, so that all available pre-prediction data contributed to the coefficient estimates.

For the final refitted model, most individual coefficients were statistically significant, and all variables represented by groups of coefficients were statistically supported as model terms. Expected counts were higher during daytime than during the night, and higher in spring and summer than in winter. Public holidays and school holidays were associated with lower expected counts. The interaction terms suggested that the time-of-day patterns were not the same for all weekdays and directions. Those estimated coefficients were then used to derive the expected normal count for each observation in the prediction dataset, which covers the period from 1 May 2025 to 30 April 2026.

5 Defining and Analyzing Deviations

After the expected normal counts had been estimated, they were compared with the observed counts in the prediction period. The comparison was made at the level of the 2-hour observation. For each 2-hour observation, the observed-minus-expected difference was computed. Based on these differences, observations were classified as deviations or non-deviations.

Since some differences in the count data are expected, an observation was classified as a deviation only if several conditions were met. First, the prediction had to be supported by a sufficient number of comparable past observations. Since the comparison was made within relatively narrow groups, defined by site, time of day, day of the week, and month, the minimum number of reference observations was set at 10. This threshold was satisfied for more than 80.5% of observations. Second, the difference between observed and expected counts had to be large in absolute terms, larger than 25. Third, it also had to be large relative to the expected count and exceed 75% of that value.

When expected counts are low, even a small difference may correspond to a large percentage change, although a difference of a few cyclists is not meaningful in practice. The absolute threshold is used to exclude such cases. When expected counts are high, the same difference represents only a small share of total traffic and does not indicate a substantial change in use. The relative threshold is used to exclude such cases. These criteria therefore ensure that only differences that are large in practical terms are classified as deviations.

After applying these criteria, 129,793 out of 1,177,575 observations in the prediction period were classified as deviations. This corresponds to 11.0% of all 2-hour observations. This proportion is used as the reference deviation share in the descriptive comparisons below. The size of the deviations ranged from -374.54 to 1889.47 cyclists.

The deviations were then separated according to the sign of the observed-minus-expected difference. There were 76,613 positive, higher-than-expected deviations and 53,180 negative, lower-than-expected deviations. Higher-than-expected deviations had a median size of 50.58, with an interquartile range from 34.82 to 83.08. Lower-than-expected deviations had a median size of -38.75, with an interquartile range from -53.40 to -30.60. Positive deviations were therefore not only more frequent, but also larger in absolute median size (see Figure 4).

The descriptive comparisons showed that the shares of the deviations differed across months, weekdays, and 2-hour intervals. The highest shares were observed in March and April. In these months, 14.6% and 16.2% of observations were classified as deviations, compared with the overall share of 11.0%, and most deviations were higher-than-expected counts. From July to October, lower-than-expected deviations became relatively more common. Thus, a seasonal pattern was present both in the occurrence of deviations and in their direction.

Deviations were also more frequent on working days than on weekends. From Monday to Friday, the deviation share ranged from 11.6% to 14.1%. On Saturday and Sunday, it was approximately 6.9%. Counts above the expected level were observed more often during the working week than

during weekends. Negative deviations followed the same direction, but the difference was less pronounced.

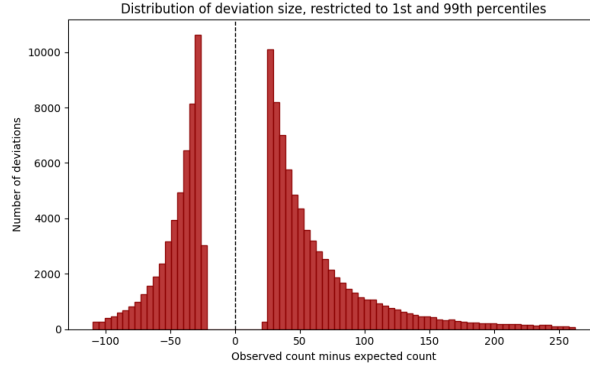


Figure 4: Distribution of deviation size, restricted to the 1st and 99th percentiles

The strongest temporal differences were observed across 2-hour intervals. Night intervals had very low deviation shares, ranging from 0.2% to 1.0%. This is partly a consequence of the absolute threshold rule used in the deviation definition. When expected and observed night-time counts are both low, even relatively large proportional differences are unlikely to exceed 25 counts. From 06:00 onward, deviations were more common. The highest shares of the deviations were observed in the afternoon, with 21.3% at 14:00 and 24.5% at 16:00. At these times, both higher- and lower-than-expected deviations were common.

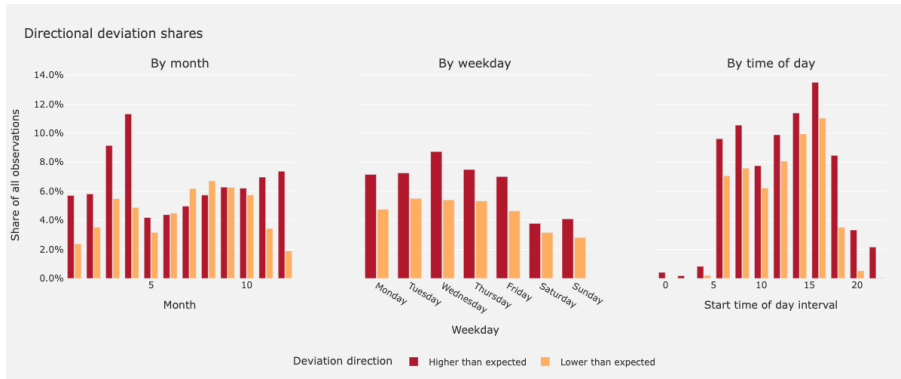


Figure 5: Directional deviation shares by month, weekday and time of day

The previous comparisons were based on the variables that were included in the expected normal count model. They are therefore considered with respect to both the regular structure in cyclist counts and the irregular structure expressed through deviations. Weather conditions and event variables, however, were not included in the normal count model. They are used here only to describe the deviation part of the count structure.

For weather conditions, precipitation and temperature were used as proxies. For precipitation, the overall deviation share was similar during dry and light precipitation periods, approximately 11.2%, and lower for moderate and heavy precipitation periods and periods with snow. The direction of deviations was different. In dry periods, deviations were more often above the expected count. With precipitation, especially moderate and heavy precipitation, deviations were more often below the expected count. Thus, precipitation appears less as a condition increasing the number of deviations and more as a condition changing their direction.

For temperature, the deviation share increased across the temperature intervals. It was 3.3% below 0°C, 13.1% between 15°C and 20°C, and about 16% above 20°C. The increase was observed for both positive and negative deviations, although positive deviations remained more frequent in most intervals (see Figure 6). For the highest temperature interval, however, the direction of deviations changed. Lower-than-expected deviations were more common than higher-than-expected deviations. This suggests that very high temperatures reduce cycling counts relative to the expected level.

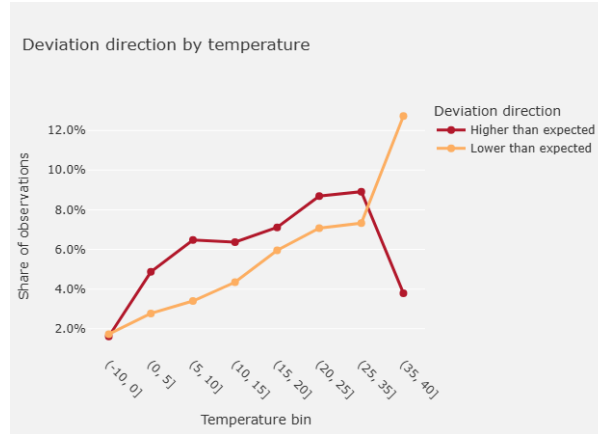
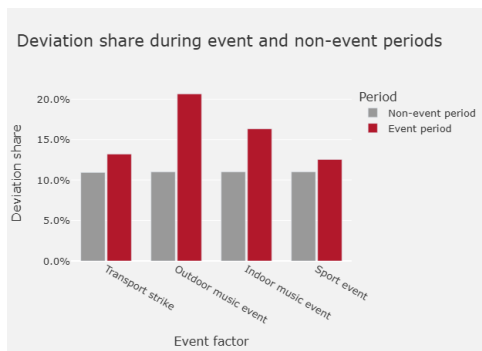
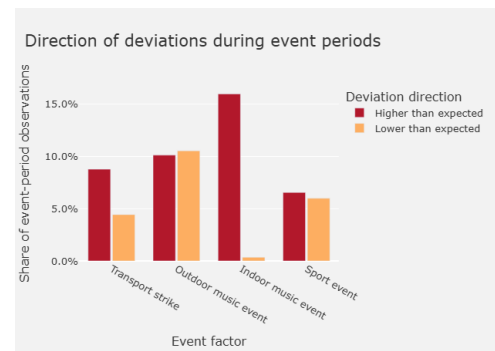


Figure 6: Share of deviations by temperature and direction

Deviation shares were also compared with respect to the presence or absence of events — transport strikes, outdoor and indoor cultural events and sport events, that can potentially disrupt cycling patterns. For all event types, the deviation share was higher during event periods. The largest difference was observed for outdoor music events, where the deviation share was 20.6%. The increase was present in both directions (see Figure 7). For indoor events, however, the higher deviation share was almost entirely due to higher-than-expected counts. Transport strikes also produced mainly higher-than-expected deviations, but the difference from non-strike periods was smaller. For sport events, the increase in the share of deviations was marginal.



(a) Deviation share during event and non-event periods



(b) Direction of deviations during event periods

Figure 7: Deviation patterns during event periods

The descriptive comparisons considered each factor separately. To assess whether differences in the occurrence of deviations remained when the factors were considered together, a regression framework was used. Three outcome variables were considered: any deviation, higher-than-expected deviation, and lower-than-expected deviation. Given that all outcomes are binary,

logistic regression was used. Besides weather and event factors, temporal covariates were also included in the model so that associations with weather and events were not attributed to temporal differences in deviation occurrence.

To fit the logistic models, a pipeline was built using the scikit-learn package. The pipeline first pre-processed the explanatory variables. Temperature was treated as a numeric variable and standardised. Precipitation category, month, weekday, and 2-hour interval were treated as categorical variables and converted into dummy variables using one-hot encoding. Event indicators were already binary and were passed to the model without transformation.

The models were trained on 70% of the data, and their performance was assessed on the remaining 30% hold-out sample. The split was stratified by the outcome variable, so the share of deviations was preserved in both parts. The mean predicted probabilities were close to the observed shares, which means that the models reproduced the overall frequency of the outcomes. The ROC AUC values were 0.79 for any deviation, 0.78 for higher-than-expected deviation, and 0.80 for lower-than-expected deviation. Average precision was lower, especially for the directional outcomes, because these outcomes were less frequent.

The logistic regression results were broadly consistent with the descriptive comparisons (see Table 2).

Table 2: Odds ratios for logistic regression models of deviation occurrence and direction

Variable	Any deviation	Higher-than-expected deviation	Lower-than-expected deviation
Temperature mean	1.35 [1.33, 1.37]	1.68 [1.65, 1.72]	0.90 [0.88, 0.91]
Light precipitation	0.82 [0.81, 0.84]	0.60 [0.59, 0.62]	1.18 [1.16, 1.22]
Moderate precipitation	0.72 [0.69, 0.75]	0.32 [0.29, 0.34]	1.40 [1.31, 1.52]
Heavy precipitation	0.93 [0.87, 0.98]	0.25 [0.22, 0.33]	1.37 [1.21, 1.71]
Snow	0.67 [0.62, 0.73]	0.47 [0.42, 0.55]	1.35 [1.11, 1.54]
Outdoor music event	2.40 [2.10, 2.78]	3.27 [2.63, 3.88]	2.32 [2.10, 2.66]
Indoor music event	2.20 [1.75, 2.73]	2.14 [1.69, 2.71]	0.93 [0.88, 1.00]
Sport event	1.68 [1.34, 2.02]	1.36 [1.04, 1.68]	1.90 [1.50, 2.28]
Transport strike	1.06 [1.03, 1.11]	1.14 [1.09, 1.19]	1.01 [0.92, 1.09]

Note. Odds ratios are exponentiated mean coefficients from 100 bootstrap samples. In each sample, observations were drawn with replacement and the full pipeline was refitted. Confidence intervals are the exponentiated 2.5th and 97.5th percentiles of the bootstrap coefficients.

The same differences remained after the variables were considered together. Precipitation and temperature were associated with opposite directions of deviation. Precipitation was related to fewer higher-than-expected deviations and more lower-than-expected deviations, while temperature showed the reverse relation. Event indicators were generally related to higher odds of deviations. This relation was present in both directions for outdoor music events, but mainly in higher-than-expected deviations for indoor music events and transport strikes. Sport events were also positively associated with deviations, but the association was weaker than for cultural events.

6 Site-Level Differences in Deviation Patterns

For systematic differences in irregular cycling counts to be identified across locations, deviations were also summarised at the site level. For each counting site, the total deviation share was computed together with the shares of higher-than-expected and lower-than-expected deviations.

Among the counting sites with the highest total deviation shares, several belonged to the same municipality or local area, such as Brasschaat, Kortrijk, Oostende, Brugge, Wervik, Heist-op-den-Berg, Geel, and Kampenhout (see Figure 8). Many of these municipalities or local areas are characterised by a stronger urban presence or by a coastal location. They are places where the observed cyclist counts may represent more than one type of movement, including work commuting, local trips, and recreational movement. A factor affecting at least one of these types of movement may therefore contribute to the observed deviations.

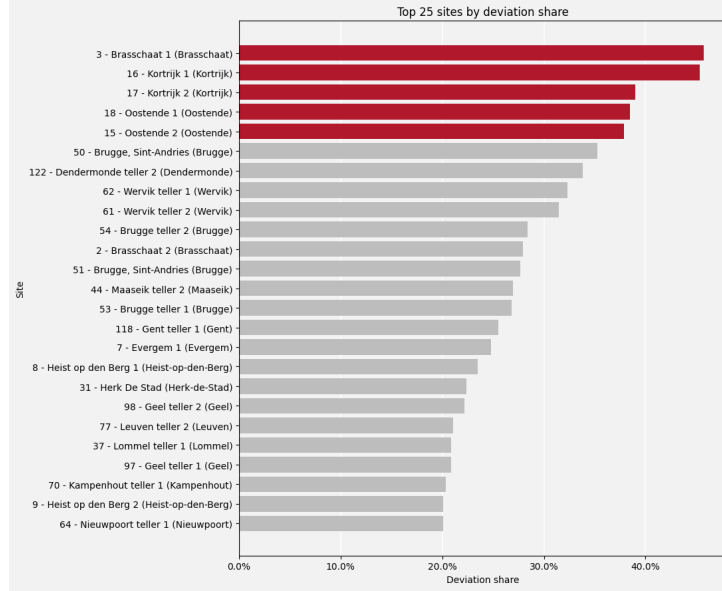


Figure 8: Top 25 sites by the share of deviations

The higher- and lower-than-expected deviations had different site-level distributions. Higher-than-expected deviations were more broadly distributed across sites (see Figure 9). They appeared in many locations, both urban and peri-urban, but only a limited number of sites had relatively high positive-deviation shares. This means that the ability of cycling infrastructure to absorb higher-than-expected traffic should be assessed and ensured across many locations. Lower-than-expected deviations were concentrated in a small number of the larger sites, where the share of such deviations was high (see Figure 9). For these sites, repeated lower-than-expected counts point to reduced use of the counted route. Restrictions in access, local works, or reduced usability of routes under external conditions such as weather may be responsible for this pattern. These factors need monitoring, and the associated sites require targeted interventions able to mitigate the impact of those factors.

For ease of monitoring and targeted intervention planning, a formal rule for site characterisation was defined. Sites with a deviation share below 10% were classified as stable, low-deviation-frequency sites. Sites with a deviation share of at least 10% were classified as unstable. For unstable sites, the relative shares of higher- and lower-than-expected deviations were compared. A site was classified as having predominantly higher-than-expected deviations when the share of observations classified as higher-than-expected deviations was more than 1.5 times the share classified as lower-than-expected deviations. The reverse rule was used for predominantly lower-than-expected deviations. Sites for which neither direction exceeded the other by this factor were classified as mixed deviation direction.

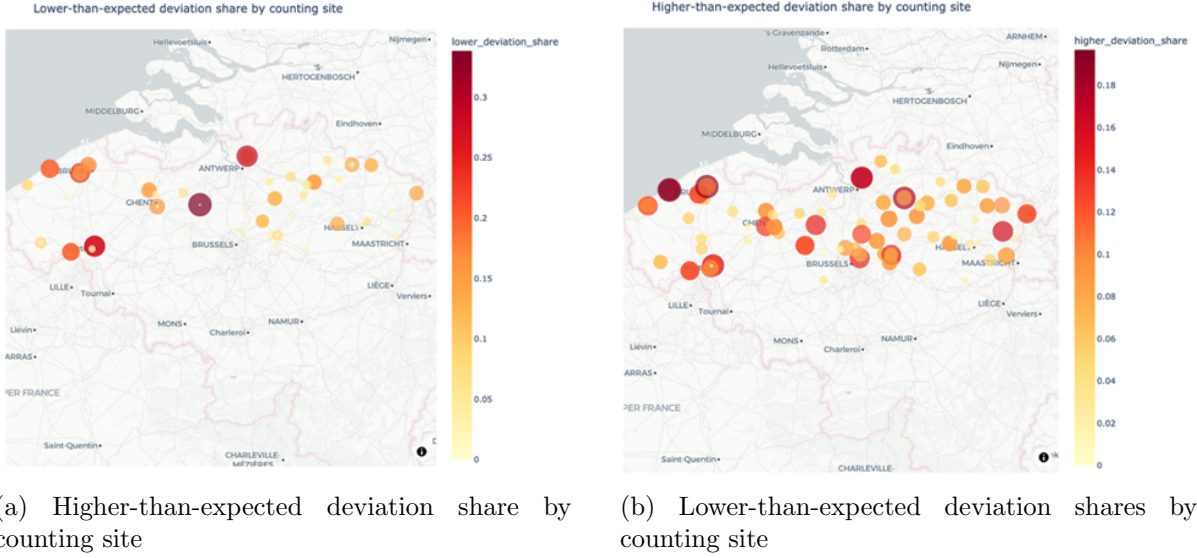


Figure 9: Higher- and lower-than-expected deviation shares by counting site

According to this rule, 78 sites were classified as stable. Among the unstable sites, 27 had predominantly higher-than-expected deviations, 14 had predominantly lower-than-expected deviations, and 17 had a mixed deviation direction (see Figure 10). Stable sites had a mean deviation share of 4.1%. For sites with predominantly higher-than-expected deviations, the mean deviation share was 13.9%. The corresponding mean shares were higher for mixed-direction sites and predominantly lower-than-expected sites, at 21.6% and 30.3%. Thus, lower-than-expected deviation profiles were less common, but, when present, they were associated with larger site-level deviation shares.

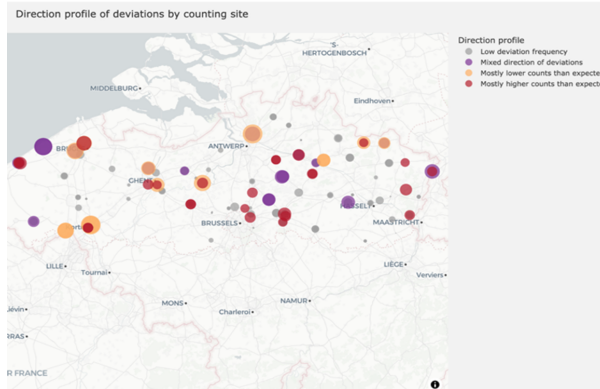


Figure 10: Direction profile of deviations by counting site

The direction profile describes whether deviations at a site are mainly positive, mainly negative, or mixed. It does not, however, show whether these deviations are tied to specific external factors. For this reason, a second characterisation was made. For each site, the deviation share under normal external conditions was first computed. Normal external conditions were defined as periods without transport strikes, cultural events, sport events, meaningful precipitation, or extreme cold or warm weather for Belgium.

For unstable sites, sensitivity to an external factor was defined by comparing the deviation share under that factor with the site-specific deviation share under normal external conditions. A site was considered sensitive to a factor when this difference was at least 5 percentage points.

The rule was applied separately to precipitation, cold weather, warm weather, transport strikes, cultural events, and sport events. If no factor satisfied this condition, the site was classified as unstable independent of the considered external factors.

Most unstable sites were factor-sensitive. Thirty-nine sites were sensitive to one factor, and 13 sites were sensitive to more than one factor. Six sites were unstable without sensitivity to any of the considered factors. Warm weather was the most common main sensitivity factor. Sensitivity to warm weather was not restricted to one location. It was observed at coastal sites, including Oostende, Brugge, and Nieuwpoort, and also at inland urban and peri-urban sites (see Figure 11). Transport-strike sensitivity had a narrower distribution. It appeared mainly at sites in municipalities with a stronger commuting context, including Gent, Leuven, and Kortrijk. Cultural-event sensitivity was limited to a few sites. It appeared mainly at Brugge and Oostende sites. Vilvoorde was the only site for which cultural events were the main sensitivity factor. Sport-event sensitivity was observed only for one site, the Dendermonde totem.

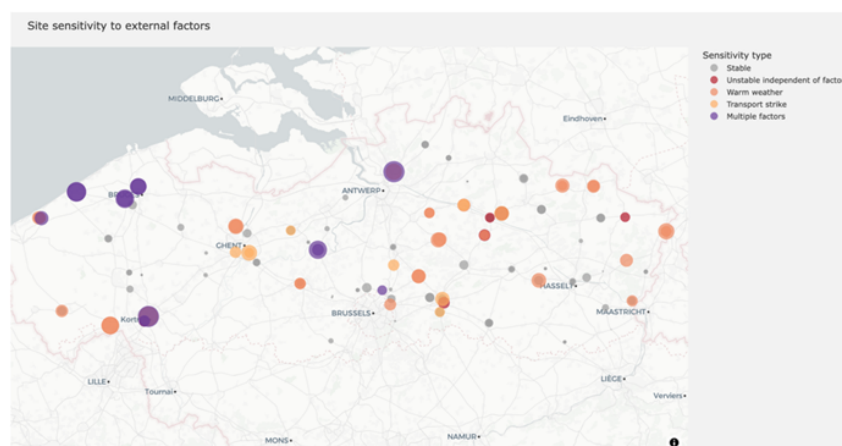


Figure 11: Site sensitivity to external factors

Sites mainly sensitive to warm weather had a mean deviation share of 18.8%, and sites mainly sensitive to transport strikes had a mean deviation share of 15.9%. Sites sensitive to multiple factors had the highest mean deviation share, 29.1%. Thus, multiple-factor sensitivity represented not only a more complex profile, but also a higher overall frequency of deviations.

The sensitivity classification gives the site categories a practical interpretation. Stable sites require only routine monitoring, since deviations were uncommon. Unstable sites independent of the considered factors require broader follow-up. These sites may either have generally higher variability in cyclist counts, or their deviations may be related to factors outside the present analysis. Such factors may include access restrictions, local works, or changes in route choice. They need to be monitored if interventions are to be targeted.

Warm-weather-sensitive sites should be monitored especially during warm periods. It needs to be ensured that infrastructure can accommodate increasing cycling counts during such periods, and that cycling does not become an unusable mode of transportation when temperatures are extreme. Transport-strike-sensitive sites should be treated as disruption-sensitive locations. Their monitoring is most relevant during announced strikes, since deviations at these sites are more likely to occur when the usual transport system is disturbed. Cultural-event and sport-event sensitivity were much more localised. These profiles are therefore less useful for general network planning, but useful for site-specific event management.

Multiple-factor-sensitive sites are the most important group for targeted intervention. They had the highest average deviation share, about 29.1%, and their deviations were not tied to

a single condition. These sites should be prioritised because they combine frequent deviations with sensitivity to more than one external factor. Their management cannot rely on one type of intervention only; it requires monitoring across several conditions, such as warm weather, events, strikes, or precipitation, as well as factors not included in the present analysis.

7 Dashboard Construction and Deployment

Since the analysis carried out in this project is complex and multi-stage, a dashboard was developed to present a simplified version of it for users such as policymakers and infrastructure planners. The final application was built using Shiny for Python and deployed through Shinyapps.io. The dashboard is divided into two main sections. The Exploratory Data Analysis section presents baseline information through five parts: Traffic Overview, Temporal Rhythm, Economics and Environmental Factors, Weather Dimensions, and Special Events Patterns. The Deviations Analysis section presents the results of the deviation analysis through five parts: Deviations Overview and Sizing, Temporal Dimensions, Weather Effect, Special Events, and Spatial Patterns.

Moving the dashboard from a local environment to a live deployment required memory optimisation, since the full dataset contains more than 4.5 million observations. To reduce the size of the deployed application, only the columns needed for visualisation were retained and saved as compressed `.parquet` files. The active memory use of the application was also reduced through data downcasting. Text variables were converted to categorical types, binary event indicators and temporal bins were stored as 8-bit integers, and continuous numeric variables were stored as 32-bit floats. The deployment was then carried out from a clean local directory containing only the application script and the required `.parquet` files, while the raw datasets were excluded from the cloud bundle. The upload was managed through the `rsconnect` CLI, with the Python version set to 3.11.5 to keep the deployment environment consistent with the local environment.

8 Discussion and Limitations

The observed seasonal variation, temporal concentration, and sensitivity to external factors indicate that cycling behaviour is dynamic rather than stable over time. This implies that both the expected-count model and the deviation classification require regular updating to remain relevant. Given that many external factors required for this type of analysis need to be manually collected, keeping the analysis up to date cannot be fully automated. Even when such information is available, it has to be matched to the count observations by date, location and, where available, time. The resulting variables therefore describe the conditions assigned to an observation, but not necessarily the exact exposure of a counting site to a given factor. Lastly, deviations are defined relative to expected counts. They therefore depend on the model used to define normal cycling traffic and on the thresholds used for classification. The results should therefore be interpreted as deviations from the modelled normal level of traffic, not as direct evidence of disruption.

Nonetheless, the analysis is still useful as a monitoring tool. It combines repeated count data, calendar structure, weather conditions, and selected event information in one framework. It therefore allows deviations to be inspected not only as isolated differences between observed and expected counts, but also in relation to the conditions under which they occur.